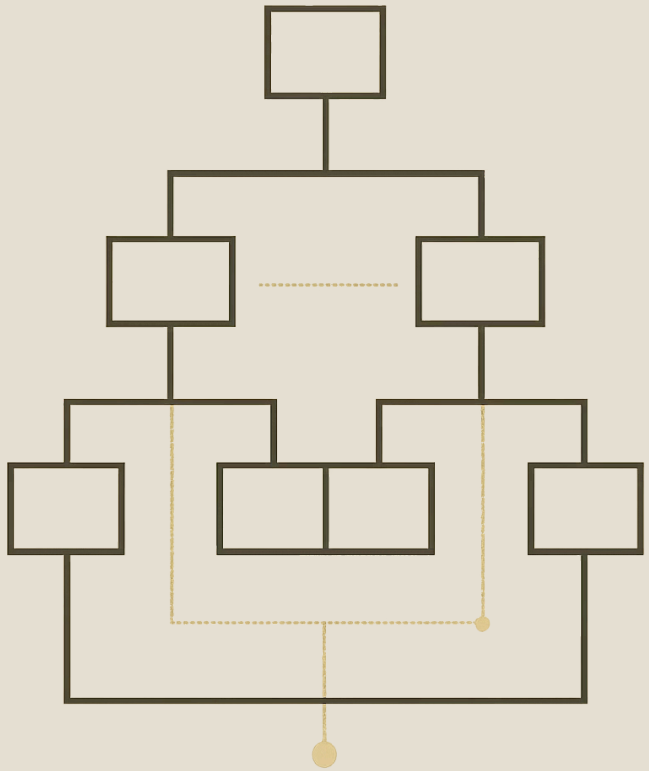

Hierarchical Clustering in Data Mining

Cluster Analysis · Unit 4 — Hierarchical Methods

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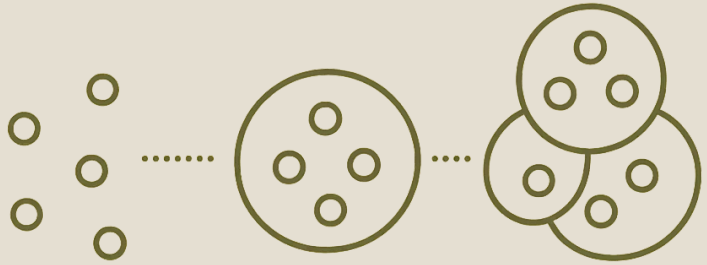


What Is Hierarchical Clustering?

Hierarchical Clustering groups similar data objects into clusters by building a **tree-like hierarchy**. Unlike partitioning methods, the number of clusters is **not fixed in advance**. Results are visualized using a **dendrogram**.

🌳 Hierarchy

Creates nested cluster relationships

📊 Flexible

No preset cluster count needed

🔬 Applications

Data mining, bioinformatics, image analysis

Two Types of Hierarchical Clustering

Agglomerative

Bottom-Up Approach

- Starts with individual data points
- Gradually merges closest clusters
- Continues until one cluster remains

Divisive

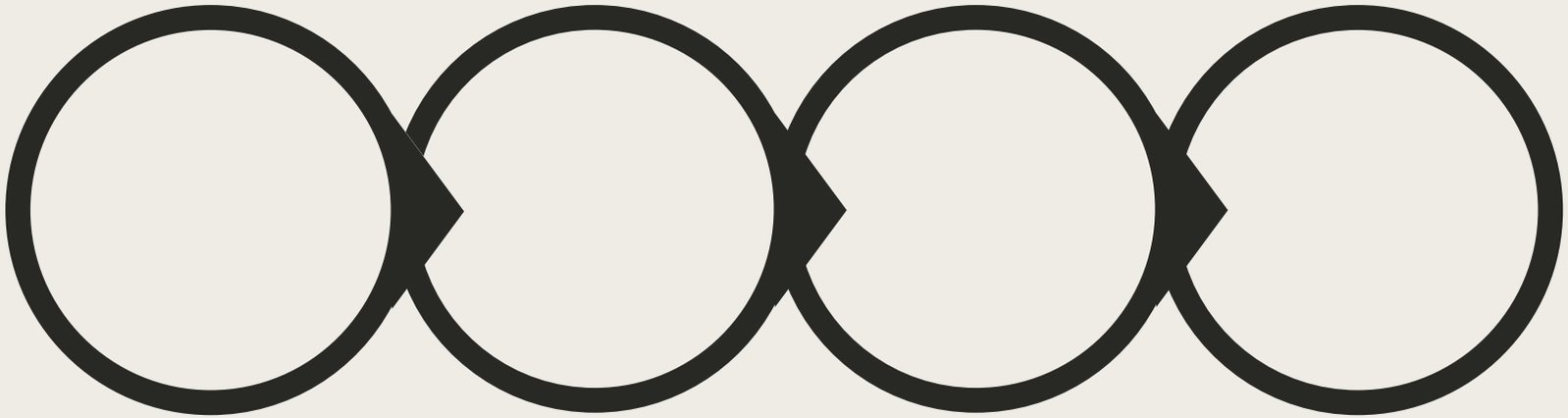
Top-Down Approach

- Starts with all points in one cluster
- Gradually splits into smaller clusters
- Continues until each point is separate

Agglomerative Clustering

MOST COMMONLY USED

Agglomerative clustering begins with each data point as its own cluster and merges the closest pairs step-by-step until a single cluster remains.

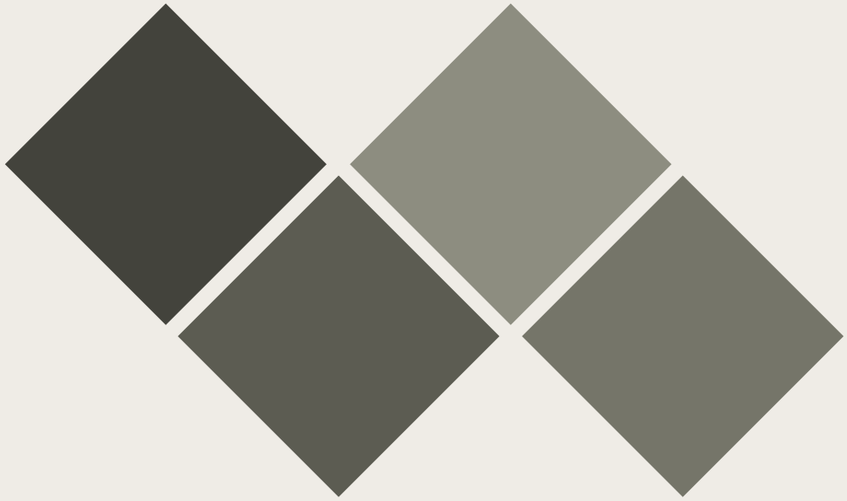


This greedy, bottom-up strategy is the most widely adopted hierarchical method due to its simplicity and effectiveness on small to medium datasets.

Divisive Clustering

TOP-DOWN APPROACH

Divisive clustering starts with all data points in one cluster and recursively splits them into smaller groups.



Divisive clustering is less common due to higher computational cost — evaluating all possible splits at each step is expensive.

Linkage Methods

Linkage methods define how the **distance between clusters** is calculated — directly affecting cluster shape and quality.

Single Linkage

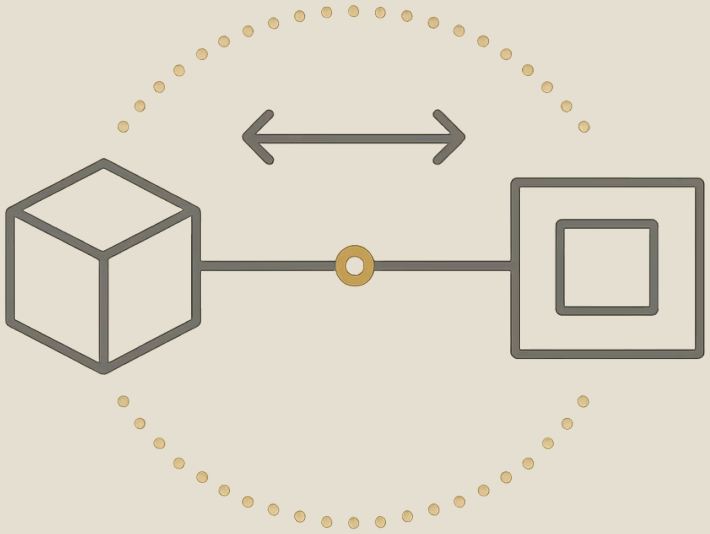
Distance between the **nearest** points of two clusters. Can produce elongated, chain-like clusters.

Complete Linkage

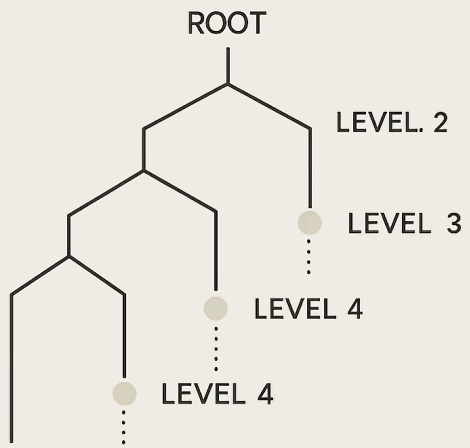
Distance between the **farthest** points. Tends to produce compact, spherical clusters.

Average Linkage

Average distance between **all point pairs**. A balanced approach between single and complete linkage.



The Dendrogram



Reading a Dendrogram

A dendrogram is a **tree-like diagram** that visualizes how clusters merge or split. The **height of each branch** represents the distance at which clusters were combined.

- **Understand relationships** between clusters
- **Select a cutoff level** to determine cluster count
- **Visualize structure** of the full hierarchy

Advantages of Hierarchical Clustering

No Predefined K

No need to specify the number of clusters before running the algorithm.

Visual & Interpretable

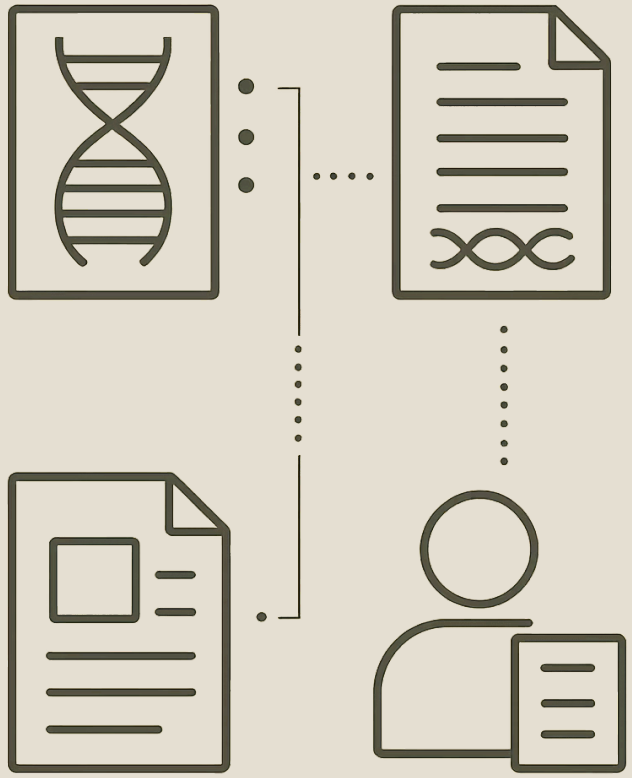
Dendrograms provide an intuitive, visual summary of cluster relationships.

Pattern Discovery

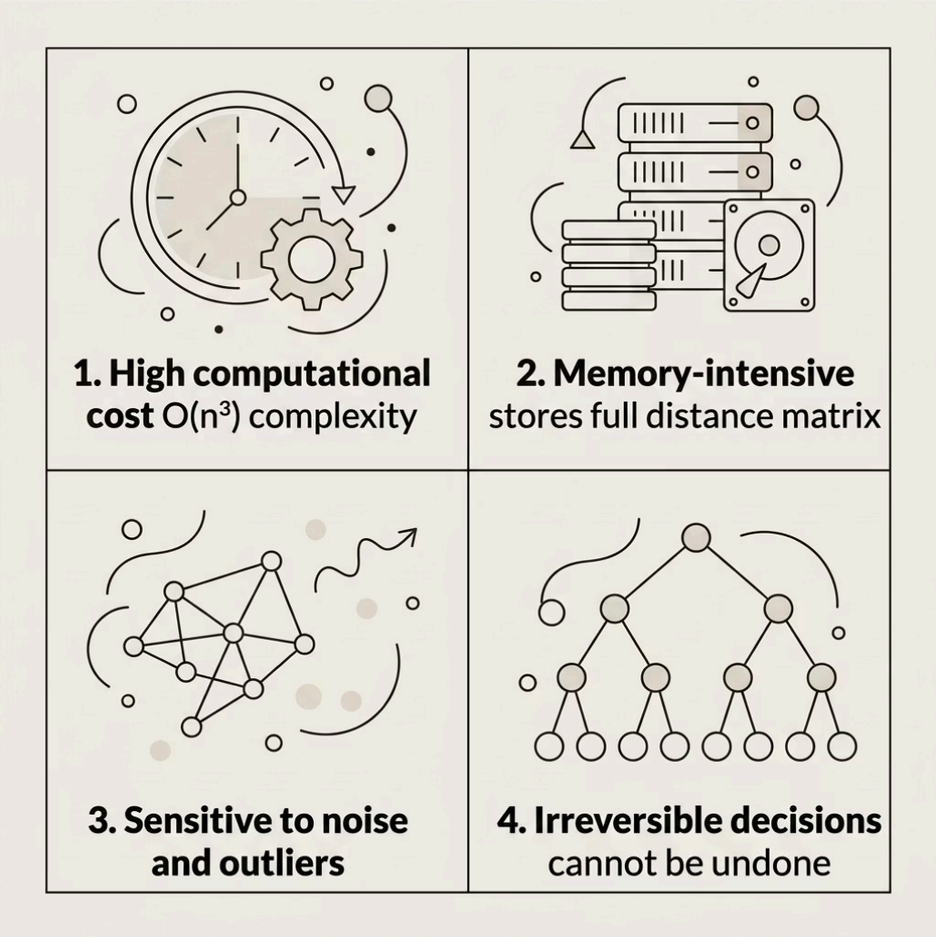
Effective at uncovering hidden structures in small to medium datasets.

Domain Versatility

Widely applied in gene analysis, document classification, and customer segmentation.



Disadvantages & Limitations



Key Takeaways

Hierarchical clustering is powerful for exploratory analysis but has real constraints at scale.

⚠️ Not for Big Data

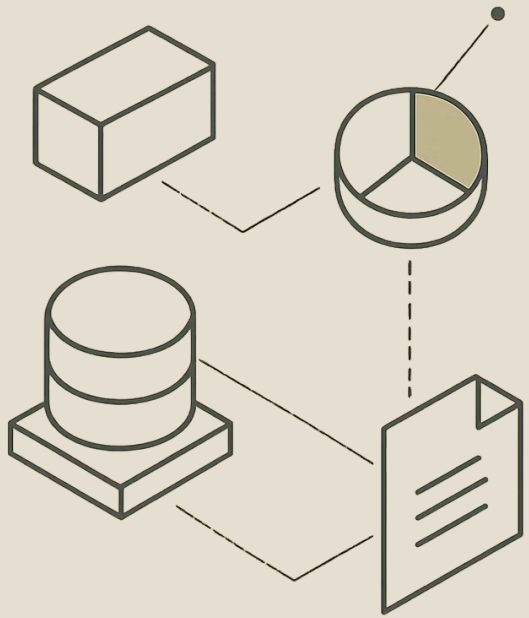
Computationally expensive — $O(n^3)$ — making it impractical for very large datasets.

⚠️ Greedy & Irreversible

Once a merge or split occurs, it cannot be undone, which may compound early errors.

⚠️ Noise Sensitivity

Outliers can significantly distort cluster boundaries and dendrogram structure.

Thank You

Questions or discussions are welcome.